



Ammonia/hydrogen non-premixed turbulent jet flames stabilized on a hot and diluted co-flow burner

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ABSTRACT

The current study investigates NH₃/H₂ non-premixed flames stabilized on a Jet-in-Hot-Co-flow (JHC) burner, with a focus on identifying the conditions that enable transition to the MILD regime. Experiments were conducted at a fixed co-flow temperature of 1180 K, with co-flow oxygen concentrations in H₂/N₂/O₂ combustion products ranging from 3 % to 9 % by mass, and hydrogen dilution in the fuel varied between 0 % and 15 % by volume. Flame structure and radical distribution were characterized using simultaneous planar laser-induced fluorescence (PLIF) imaging of NH and OH radicals, in combination with temperature measurements. The results show that a small amount of hydrogen (1 % vol) effectively reduces the local auto-ignition temperature below the co-flow temperature, independent of the oxygen concentration. This implies that minor initial dissociation of ammonia can help in the transition to the MILD combustion regime. At 9 % oxygen concentration, three distinct combustion regimes were observed as hydrogen dilution increases from 0 % to 15%: no visible reaction, a stable visible reaction zone, and a standard turbulent flame. In contrast, such regime transitions were absent at 3 % oxygen, highlighting the essential role of oxidizer reactivity in enabling MILD combustion. Despite meeting the temperature-based criteria for MILD combustion, NO emissions remained relatively high and consistent (~800 ppm) at 250 mm downstream of the jet across all MILD cases. N₂O exhibited strong sensitivity to hydrogen dilution, increasing from 90 ppm to 780 ppm as the hydrogen dilution rose from 3 % to 15 %. These findings highlight the need for revised and more comprehensive criteria to define MILD combustion in NH₃-fueled environments.

Novelty and significance statement

This study reports the first experimental investigation of the transition boundary between MILD and conventional regimes in non-premixed NH₃/H₂ flames using a Jet-in-Hot-Co-flow burner. The results demonstrate that even limited NH₃ dissociation markedly lowers the auto-ignition temperature and thereby promotes MILD stabilization. Moreover, the work shows that temperature-based criteria alone are inadequate to capture the low-emission characteristics of MILD, highlighting the need for a revised and more comprehensive definition tailored to ammonia-fueled systems.

1. Introduction

To achieve carbon neutrality and a sustainable energy future, development and deployment of renewable energy sources are essential. Among various alternatives, hydrogen (H₂) has emerged as a promising candidate, as it avoids the carbon dioxide emission problem. However, its widespread application faces significant challenges, primarily stemming from its low volumetric energy density under ambient conditions (3 Wh/L) and wide flammability limits, which both complicate processes such as liquefaction, compression, and storage [1,2].

In contrast, ammonia (NH₃) has attracted growing interest as an efficient hydrogen carrier [3], containing 17.8 % hydrogen by weight and exhibiting a relatively high gravimetric energy density of 22.5 MJ/kg [4]. Moreover, ammonia benefits from a well-established global infrastructure for storage and transportation, making it a viable

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candidate for integration into low-carbon energy systems [5,6]. Despite these advantages, ammonia itself is inherently toxic and corrosive, and its combustion characteristics differ significantly from those of conventional hydrocarbon fuels: poor flame stability, a higher ignition energy threshold, and increased nitrogen oxide (NO_x) emissions [7]. Of particular concern is nitrous oxide (N_2O), a potent greenhouse gas with a global warming potential approximately 300 times greater than carbon dioxide (CO_2) [8,9]. The mitigation of such emissions often necessitates cost-intensive post-combustion treatment technologies to comply with increasingly stringent environmental regulations.

To address the unfavorable combustion properties of NH_3 , various enhancement strategies have been proposed, such as blending with more reactive fuels [10–12], increasing initial temperature or pressure [13, 14], and oxygen enrichment [15–17]. Among these, fuel blending, especially with hydrogen, has proven particularly effective in improving flame stability and reactivity [10,18,19]. For example, Zhang et al. [17] utilized detailed chemical kinetic modeling to show that hydrogen addition and oxidizer enrichment significantly increase laminar flame speed and reduce ignition delay times in NH_3/H_2 mixtures. Similar conclusions have been drawn by other researchers [10,20]. He et al. [21] demonstrated that the addition of just 1 % H_2 (by vol.) to ammonia reduced the ignition delay time by a factor of three at 40 bar, with further improvements at higher pressures. Experimental studies such as that of Choi et al. [22] investigated non-premixed NH_3/H_2 /air counterflow flames and found that hydrogen addition extended flame blowout limits, increased radical concentrations (H , OH , O), and raised the maximum flame temperature. However, the benefits of hydrogen blending sometimes come at the cost of increased NO_x emissions. Several studies have reported that elevated concentrations of OH and O radicals in NH_3/H_2 flames can promote NO and N_2O formation, resulting in higher NO_x emissions compared to pure NH_3 combustion [23,24].

Moderate or Intense Low-Oxygen Dilution (MILD) combustion has emerged as a promising strategy to address the limitations [25–27]. Under this combustion regime, the inlet temperature of the reactant mixture exceeds its auto-ignition temperature, while the maximum temperature rise during combustion remains lower than the initial temperature of the mixture. The former condition enables distributed auto-ignition, which facilitates the development of a uniform temperature field and mitigates the risk of flame extinction. While the latter condition limits peak combustion temperatures, typically below 1600 K, thereby effectively suppressing the formation of thermal nitrogen oxides (NO_x). Thus, MILD combustion typically occurs in a distributed volumetric mode under an environment of high temperature and strongly diluted conditions [28,29].

Recent studies have demonstrated the effectiveness of MILD combustion in ammonia-fueled systems. Kuang et al. [30] conducted numerical investigations on various fuel blends (NH_3 , H_2 , CH_4) stabilized on a Jet-in-Hot-Co-flow (JHC) burner under different combustion regimes, confirming that MILD conditions lead to a substantial reduction in NO emissions. Ferrarotti et al. [31] examined NH_3/H_2 combustion in a furnace operating under MILD conditions and identified an optimal balance between NO_x formation and ammonia slip at an equivalence ratio near stoichiometric conditions ($\phi \approx 0.95$). In a cyclonic burner configuration, Sorrentino et al. [32] experimentally validated the feasibility of pure ammonia combustion under MILD conditions, emphasizing the necessity of maintaining temperatures above 1300 K to achieve stable combustion and minimal pollutant emissions.

One of the key parameters governing the realization of MILD combustion in fuel blends is the set of conditions that facilitate its establishment [33]. Among these, the desired dilution level, oxidizer temperature, and composition play particularly critical roles. To investigate their combined effects, zero-dimensional reactor models, such as the perfectly stirred reactor (PSR) also known as the well-stirred reactor (WSR) have been widely employed to characterize ignition behavior and reaction regimes under controlled thermal and chemical environments. Luan et al. [25] utilized a PSR model to examine $\text{CH}_4/\text{O}_2/\text{N}_2$ mixtures (τ

$= 1.0$ s, $\Phi = 1.0$) within the MILD combustion regime proposed by de Joannon [28] and Cavaliere [29], focusing on the influence of critical oxygen concentration and heat loss ratio (HLR). Their results showed that the minimum oxygen concentration required to sustain MILD combustion increased from 9.7 % under adiabatic conditions to 18 % and 29.5 % at HLRs of 25 % and 50 %, respectively. These findings underscore the need to couple elevated oxidizer temperatures with substantial oxygen dilution to overcome thermal losses. Moreover, CO_2 was found to be more effective than N_2 in promoting the transition from high-temperature combustion to the MILD regime, attributed to its superior heat capacity. Building on this modeling approach, Lewandowski et al. [33] investigated the MILD regime boundaries for binary blends for ammonia hydrogen and methane using a $T_{\text{in}}\text{-X}_{\text{O}_2}$ mapping strategy, with NO_x emissions incorporated as a performance metric. Their study revealed that although satisfying the temperature-based criterion could suggest a transition toward MILD-like behavior, this alone did not guarantee low NO_x formation, particularly for ammonia-containing mixtures. To limit NO_x concentrations below 100 ppm under stoichiometric conditions, further oxygen dilution to levels as low as 2–6 % was required, depending on the fuel composition.

It is important to note that the above studies relied exclusively on idealized reactor simulations and lacked experimental validation under practical burner conditions. To address this gap, the present study experimentally investigates the combustion characteristics of unconfined ammonia and ammonia-hydrogen blends jet flames undergoing transition to MILD combustion on a JHC burner. The effects of H_2 dilution in the fuel jet are examined. Detailed temperature profiles, as well as simultaneous measurements of NH - and OH -PLIF (Planar Laser-Induced Fluorescence) were used to shed light on fuel dissociation and the reaction zone location and structure. These three indicators were then used to imply combustion regime identification, followed by quantitative measurements of NO and N_2O under the MILD conditions. This comprehensive study is expected to provide new experimental insights into the turbulent MILD combustion behavior of NH_3/H_2 flames in hot and diluted co-flow environments, contributing to the broader understanding required for deploying ammonia-based low-carbon combustion technologies.

2. Experimental methods

2.1. Experimental setup

Experiments were conducted on a JHC burner operating at atmospheric pressure. Detailed information about the burner is available in previous studies [34,35]. Thus, only a brief description is given here. As shown in Fig. 1, the JHC burner consists of a 0.28 m long stainless-steel tube supplying the fuel blends. The tube is surrounded by a 0.10-m diameter streaming co-flow containing hot products of $\text{H}_2/\text{N}_2/\text{air}$

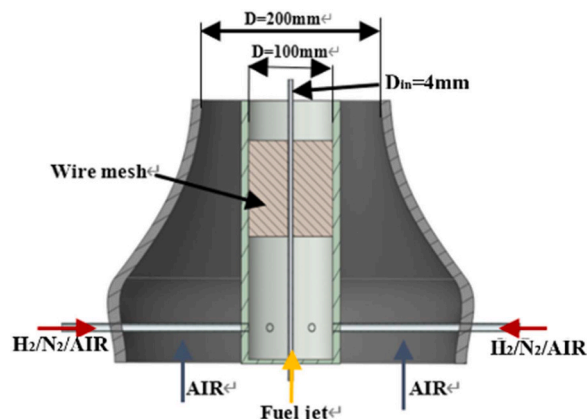


Fig. 1. Schematic illustration of the JHC burner assembly.

flames. The co-flow was conditioned by a combination of three cross-stacked screens, and a graded ceramic bead pack (5/3/2 mm, total length 100 mm) located ~ 100 mm upstream of the exit to suppress large-scale structures and ensure a uniform outflow. The inner diameter of the jet tube, denoted as D_{in} , is 4.00 mm. The burner is mounted in a wind tunnel running into atmospheric air from a fan. The exit plane of the jet is 50 mm above the outlet plane of the air contractor, enabling easy sampling and optical access. The contraction and burner assembly are mounted on a 3D transverse stage, allowing for precise adjustments in three spatial axes and providing convenient access to conduct the optical and temperature measurements.

Experiments were first conducted using pure ammonia as the baseline fuel. Hydrogen was blended with ammonia, up to 15 % by volume, to investigate the impact of dilution on flames' burning characteristics, structure, and emissions. In particular, the impact of hydrogen addition to ammonia on the establishment of the conditions of MILD combustion and its impact on pollution emission are our focus.

The co-flow stream consisted of hot products of an H_2 – N_2 -air mixture to simulate the products of ammonia combustion in any practical combustor. Nitrogen was used to control the excess O_2 concentrations in the co-flow products. The temperature of the co-flow stream at the exit plane of the annulus (T_{cof}) was fixed at ~ 1180 K for all flames.

On the co-flow side, H_2 (>99.99 %) and N_2 were regulated using Brooks mass flow controllers (SLA5851) with an accuracy of ± 0.9 % of the set point over 20–100 % full scale, while air was supplied through an identical unit specified to ± 1.0 % of reading.

On the jet side, H_2 was delivered through two Brooks controllers (SLA5851, 0–50 slm; SLA5850, 0–2000 scfm) to accommodate different enrichment levels, both with an accuracy of ± 0.9 % of the set point. The NH_3 (>99.98 %) flow was controlled by a Bronkhorst unit (F-202AV) with an accuracy of ± 0.5 % of reading plus ± 0.1 % full scale. All flow controllers were calibrated against a flow calibrator (MesaLabs Definer 220), ensuring overall flow accuracy within ± 1.0 %.

Table 1 summarizes all the experimental conditions investigated. Case names are labeled as $AxxOy$, where Axx indicates the ammonia volume percentage in the fuel (with the remainder H_2), and Oy is the O_2 mass fraction in the hot co-flow. YO_2 , YH_2O , and YN_2 denote the mass fractions of O_2 , H_2O , and N_2 in the co-flow, respectively. The fuel jet Reynolds number was fixed at $Re = 10,000$ for all cases, corresponding to jet velocities ranging from 43.75 m/s (A100) to 83.76 m/s (A85), depending on the fuel composition and resulting gas density. The excess O_2 mass fraction in the hot co-flow was set to 9 %, 6 %, and 3 %, which is consistent with the previous study [34]. Both the co-flow through the annulus and the surrounding tunnel air had a fixed velocity of 2.0 m/s in all experiments. Under these conditions, the Reynold number in the co-flow range between 1285 and 1336, indicating laminar flow regime. The use of a wind-tunnel environment ensures a stable, well-defined inflow boundary and controlled entrainment, which is advantageous for quantitative diagnostics and reproducibility.

The present burner and operating conditions differ from the Medwell/Adelaide JHC configurations in several respects, including smaller jet diameter, altered co-flow annulus geometry, and the imposition of a uniform external flow. These differences lead to modest variations in

non-dimensional parameters (Re_{jet} , Re_{cof} , and U_{cof}/U_{jet}). Such differences may influence entrainment rates and stabilization distances, but the overall trends in flame structure, temperature distribution, and pollutant formation, particularly the effects of T_{cof} and H_2 dilution, are expected to remain consistent with the underlying MILD combustion behavior.

2.2. Temperature profiles and flame appearance

Temperature profiles were measured using an Omega S-type thermocouple (Pt/10 % Rh-Pt) with a wire diameter of 125 μm along both the central axis and the radial direction of the flame. Prior to the experiments, the thermocouple was calibrated against a reference probe in a controlled furnace within the manufacturer's specifications. The recorded temperatures were corrected to account for heat transfer due to radiation and conduction between the thermocouple beads and their surroundings. Conduction losses were minimized by employing a fine-wire, bare-bead configuration with an exposed length significantly greater than the bead diameter, providing high thermal resistance to the support. Radiation losses were inherently reduced by the hot, oxygen-diluted co-flow environment, where the surrounding gas temperature was comparable to that of the bead, resulting in a small radiative temperature difference and correspondingly low radiative flux. Based on the calibration and heat transfer analysis, the uncertainty of the corrected thermocouple measurements is estimated to be ± 2 %.

The flame appearance was captured using NIKON D700 DSLR camera, utilizing an aperture setting of $f/3.2$ and an ISO value of 320.

2.3. PLIF of NH and OH

The experimental setup and optical arrangement are shown in Fig. 2. A nanosecond Nd:YAG laser (Continuum, Powerlite DLS90010) pumping a tunable dye laser (Continuum, ND6000) at 532 nm with a repetition rate of 10 Hz was used, which is the same as that used in [7]. Using Rhodamine 610–640 (3:1) dye mixture, the dye laser output was tuned to ~ 608 nm, then frequency doubled to obtain ~ 303 nm to probe the ($A^3\Pi - X^3\Sigma^-$) NH system in the (0–1) vibration band. The resulting maximum energy output of the laser was ~ 9 mJ/pulse at ~ 303 nm, which was within the linear regime for PLIF measurements.

The OH radical was captured using the same tunable dye laser and Nd:YAG system pumping at 532 nm described above. The only difference was replacing the dye with Rhodamine 6 G, enabling a maximum pulse energy of 100 mJ at ~ 566 nm. The output beam was frequency-doubled using a KDP crystal to produce UV radiation at 283.297 nm, resonant with the $Q_1(6)$ transition of the $A^2\Sigma^+ \leftarrow X^2\Pi$ (0–0) OH band. Measurements were conducted in the linear PLIF regime with a laser energy of ~ 5 mJ/pulse.

OH and NH laser beams were overlapped using a beam splitter and converted into a ~ 30 mm tall, ~ 800 μm thick laser sheet using sheet optics, and positioned above the JHC burner centerline to image the two species. The two sets of lasers and cameras were synchronized using a time delay generator, and the time delay between both laser pulses was set to 100 ns to avoid cross-talks between NH and OH fluorescence

Table 1
Experimental conditions.

Fuel jet ($Re=10,000$) NH_3/H_2 blends				Hot diluted co-flow			
Case name	H_2 dilution (vol.)	H_2 dilution (mass)	T (K)	T (K)	YO_2 (% mass)	YH_2O (% mass)	YN_2 (% mass)
A100O3, A99O3, ... A85O3	0,	0,	325	1180	3.00	10.48	86.52
A100O6, A99O6, ... A85O6	1 %,	0.12 %, 0.36 %, 0.88 %, 1.44 %, 7 %, 11 %, 15 %	325	1180	6.00	10.23	83.77
A100O9, A99O9, ... A85O9	3 %,	2.05 %	325	1180	9.00	10.12	80.88

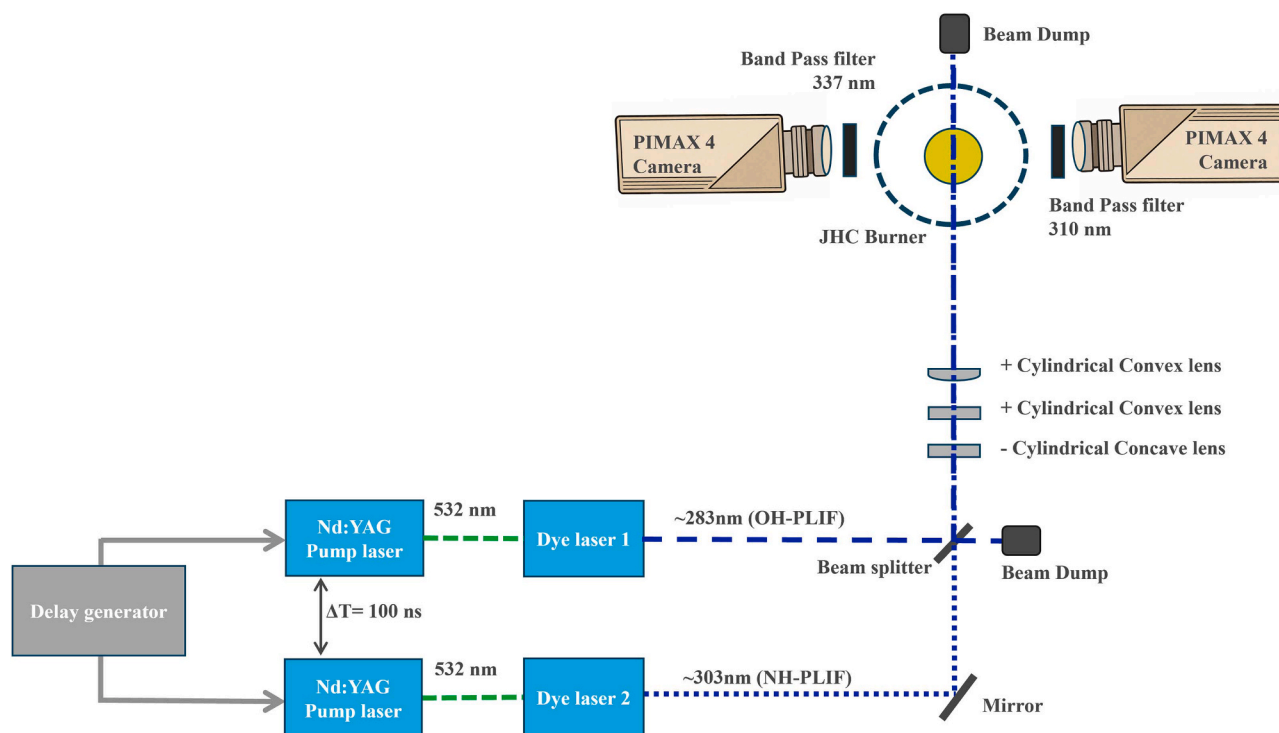


Fig. 2. Sketch of laser setup for NH-PLIF and OH-PLIF measurements.

signals.

Resonant fluorescence signals from NH and OH radicals were recorded perpendicular to the laser sheet through two image-intensified charge-coupled cameras (ICCD) (PI-MAX4, 1024×1024 pixels, Princeton Instruments) equipped with a UV objective (Cercor, $f = 100$ mm, $f/2.8$).

In the case of NH, the camera was equipped with a 10-nm bandpass filter centered at 337 nm (Edmund Optics, 65–189) to detect the emission from the $A^3\Pi - X^3\Sigma - (0, 0)$ system. The OH-PLIF fluorescence was collected using a 310 ± 10 nm band pass filter (Edmund Optics, 34–980).

For each operating condition, 400 single-shot images were recorded using a gate width of 70 ns. The raw images were background-corrected to get the final fluorescence images. It is important to note that while the line chosen in this study is less impacted by quenching, the data presented here should be regarded as qualitative. The trends, however, are semi-quantitative due to the homogeneous mixture in the co-flow and the limited range of temperatures where both radicals are present.

2.4. NO and N₂O emissions

Axial quantitative measurements of NO and N₂O were performed using a Nicolet iS10 FTIR spectrometer (Thermo Fisher Scientific), equipped with a KBr beam splitter, Globar mid-IR source, and DLaTGS detector. The system operated in the $550\text{--}4000$ cm^{-1} spectral range at a resolution of 0.5 cm^{-1} , sufficient to resolve the ν_1 band of NO and the ν_3 band of N₂O without saturation.

Combustion gases were sampled using a 0.5 mm quartz probe and transferred through a 1.5 m heated stainless-steel line (maintained above 374 K) into a 2.0 m optical path gas cell (0.2 L, ZnSe windows). The sample temperature at the cell inlet was approximately 374 K, sufficiently high to avoid condensation and minimize adsorption losses. Prior to each measurement, the cell was evacuated ($\sim 10^{-3}$ mbar) and refilled to 0.97 bar, with pressure monitored by a Baratron gauge. Background spectra were collected using high-purity N₂ at 1.1 bar.

Each spectrum was processed with zero-fitting and Boxcar

apodization. Species concentrations were extracted by fitting the resulting absorbance spectra with optimized Voigt profiles, ensuring that residuals between the measured and calculated absorbance remained below 8 %. The overall uncertainty in the NO and N₂O mole fractions was estimated to be within 10 %.

3. Results and discussion

This section presents and discusses the results of this study, with a specific focus on the combustion characteristics of the various flames and the transition to MILD combustion. The analysis begins with a description of the flame appearance, followed by an evaluation of temperature profiles. The effects of hydrogen dilution on flame structure are further analyzed using simultaneously imaged NH-PLIF and OH-PLIF. Based on these observations, the temperature-defined combustion regime is identified. Finally, measurements of NO and N₂O emissions under MILD conditions are reported and discussed.

3.1. Flame appearances

Fig. 3 presents broadband images showing the appearance of all flames investigated in this study. A camera exposure time of 1 s was used to ensure that both stable and transient combustion behaviors could be captured. The recorded flame luminosity in these images arises from spontaneous emission of hot intermediate species and radicals within the flames. In the vertical direction, only minor changes in flame appearance are observed as the co-flow oxygen concentration increases from 3 % to 9 %. In the horizontal direction, as the level of hydrogen dilution in the jet increases, the overall flame color shifts gradually toward yellow, which is attributed to the relative contributions of NO₂* and NH₂* emissions [36,20]. With H₂ addition, the strengthened H/OH radical pool and elevated local temperature promote NH₂* emission in the visible range, consistent with the observed yellowing.

Taking the 9 % O₂ case as an example, the flame height is significantly greater for the case of A100 (no hydrogen dilution) compared to the hydrogen-diluted cases. This expected behavior relates to slow

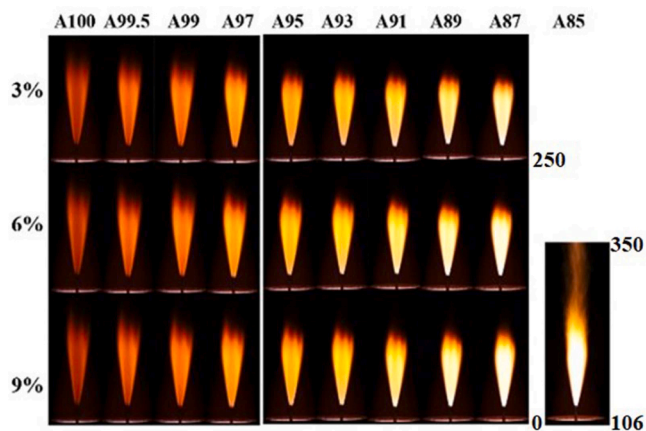


Fig. 3. Photographs of all flames at co-flow oxygen concentrations of 3 %, 6 %, and 9 %, for various hydrogen dilution levels (0–15 % H_2 by volume). Except for A85 at 9 % O_2 (image height 350 mm), each image corresponds to a physical width of 106 mm and a height of 250 mm.

reaction rate of pure ammonia to achieve complete oxidation. Furthermore, the flame root moves upstream with increasing hydrogen dilution, consistent with hydrogen's lower auto-ignition temperature and higher laminar flame speed. Notably, when 15 % hydrogen is added under 9 % O_2 co-flow conditions (case A85O9), visible flame "tails" appear, suggesting the onset of a transitional combustion flame. With further hydrogen enrichment beyond this level, the flame exhibits many features of a standard turbulent non-premixed flame appearance.

It should be emphasized that the flame photographs are qualitative, and quantitative attribution of the observed color change would require spectrally resolved or band-pass imaging.

3.2. Temperature profiles

Fig. 4 presents axial and radial temperature profiles at $H_{AB} = 10$ mm, 50 mm, and 90 mm for jet flames containing <1 % H_2 addition, under varying co-flow oxygen concentrations. The first radial profile temperature measurement was taken at $H_{AB} = 10$ mm above the jet exit, where all cases show a similar initial temperature of approximately 380 K. A noticeable temperature rise occurs around the axial position of $H_{AB} = 40$ mm, indicating the onset of intense chemical reaction and heat release. Thus, beyond this point ($H_{AB} > 40$ mm), the axial temperature distribution reflects the established combustion characteristics of the NH_3/H_2 jet flames.

At measurement heights of 10 mm and 50 mm, the radial temperature profile across the co-flow is nearly uniform, suggesting effective preconditioning and mixing of the oxidizer. This is particularly relevant given the correlation between co-flow temperature and OH radical distribution (with OH playing a critical role in NH_3 oxidation). Notably, at the lowest oxygen concentration of 3 % in the co-flow, no distinct temperature peak is observed at the axial location of $H_{AB} = 90$ mm, despite the visible presence of a flame. This behavior indicates a distributed combustion regime, in which heat release occurs gradually without the sharp temperature peak due to the low-oxygen, high-temperature environment. Additionally, spatial averaging and the thermal inertia of the thermocouple may contribute to the smoothing of localized temperature gradients.

Higher oxygen concentrations result in elevated axial temperature profiles. For the A99 case (1 % H_2 dilution), the axial temperature at $H_{AB} = 90$ mm decreases from 994 K to 897 K as the co-flow oxygen concentration is reduced from 9 % to 3 %. In contrast, the temperature profiles are nearly identical for cases with H_2 dilution <1 % under fixed co-flow oxygen conditions, suggesting that the combustion characteristics are primarily governed by fuel-side reactivity. Due to the limited thermal input from 1 % H_2 enrichment, the overall heat release remains

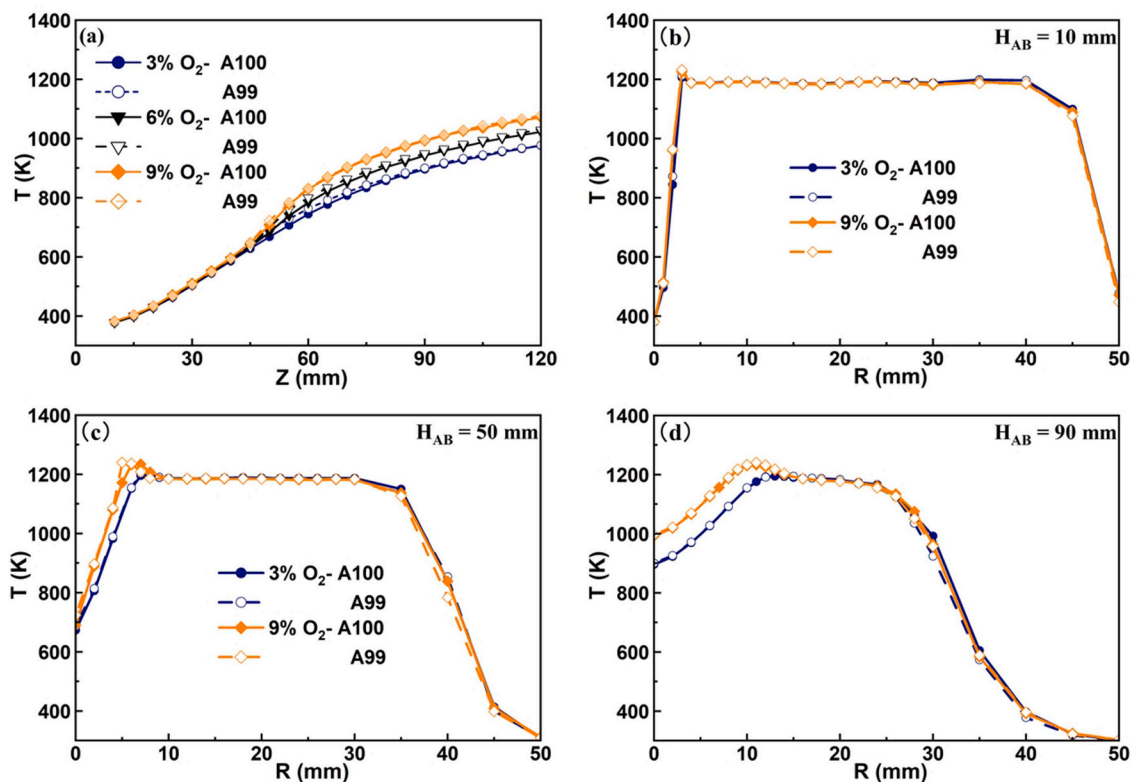


Fig. 4. Axial and radial temperature profiles for jet flames with H_2 dilution <1 % (A100–A99), under co-flow O_2 concentrations of 3 %, 6 %, and 9 %. The uncertainty of the corrected thermocouple measurements is ± 2 %.

nearly unchanged, yielding temperature profiles that closely resemble those of the pure ammonia flames.

Fig. 5 shows the temperature profiles with higher hydrogen dilution in the jet ($>1\%$ H_2 dilution by volume). Increasing the hydrogen fraction in the fuel enhances reactivity and lowers auto-ignition temperature, which helps to stabilize the NH_3/H_2 flames. At a given co-flow O_2 level, the axial temperature rise becomes more pronounced as the H_2 fraction increases (i.e., as the NH_3/H_2 ratio decreases).

However, under the very low-oxygen conditions (3% O_2), increasing the hydrogen content (up to 11% H_2 by volume, case A89O3) yields negligible changes in the flame temperature profile. This suggests that, in such oxygen-limited cases, the combustion behavior is primarily constrained by the lack of oxidizer rather than by the added fuel reactivity. In other words, at 3% O_2 , the oxygen acts as the limiting reactant, which significantly suppresses heat release even when a more reactive fuel component is added. Besides, at locations above 50 mm , the influence of excess oxygen concentration on the radial temperature distribution becomes more evident, irrespective of the NH_3/H_2 ratio. This indicates that the decomposition of NH_3 and subsequent oxidation reactions play a dominant role in heat release, as also suggested by Zhu [36]. By contrast, at $H_{AB} = 90\text{ mm}$, the thermal influence of the hot co-flow is no longer detectable, and the flame structure at this height is governed by the combined effect of turbulence and entrainment of ambient air. Moreover, as the hydrogen fraction in the fuel increases (at the same jet $Re = 10,000$), the reaction zone shifts closer to the jet centerline, likely due to changes in fuel density and jet momentum associated with adding hydrogen.

3.3. NH and OH radical planar distribution

Fig. 6 presents time-averaged PLIF images of OH and NH radicals, obtained at varying ammonia volume fractions from 100% to 85% (A100-A85) and co-flow oxygen concentrations of 3% (top row) and 9% (bottom row). It should be noted that both NH-PLIF and OH-PLIF

measurements are interpreted in a semi-quantitative manner, as collisional quenching and line-of sight absorption can influence the signal intensity [37]. In the present JHC configuration, the OH distribution in the co-flow is relatively homogeneous and the reaction zone is thin, such that absorption effects are spatially uniform and therefore negligible [38]. OH-PLIF is therefore considered to reflect relative radical distributions. In contrast, NH-PLIF is more strongly affected by collisional quenching and may also be influenced by weak self-absorption and spectral interference from OH and H_2O emissions. Accordingly, NH-PLIF and OH-PLIF are used to indicate the reaction zone and relative spatial trends.

In each image, the left half displays OH radical distributions, representing combustion activity contributed by both the co-flow and the jet flame, while the right half shows NH radicals, which indicate intermediate species generated during the thermal decomposition and partial oxidation of ammonia.

In this study, PLIF measurements for a flame with 15% H_2 dilution in the jet under 9% O_2 in the co-flow (case A85O9) were conducted up to an axial height of 150 mm . For other conditions, measurements were limited to 125 mm , as the thermal effect of the high-temperature co-flow diminishes beyond 100 mm , and the combustion process is greatly affected by the entrainment of outer air, and the MILD regime can no longer be sustained. Both NH and OH signals effectively mark the heat release of the non-premixed jet flames, with the NH-PLIF signal consistently located closer to the fuel side, in agreement with observations by Wang et al. [39]. Furthermore, the OH-PLIF intensity remains nearly uniform in the radial direction, consistent with the temperature profiles discussed in Section 3.1.

Fig. 6 reveals three distinct zones: (i) non-reacting, (ii) visible reaction zone, and (iii) standard turbulent flame. In the A100 case (pure ammonia), the absence of a detectable NH-PLIF signal indicates a non-reacting regime. As hydrogen addition increases from 1% to 15% , NH signal intensity gradually appears and intensifies, indicating the onset and enhancement of chemical activity. Under the A85O9

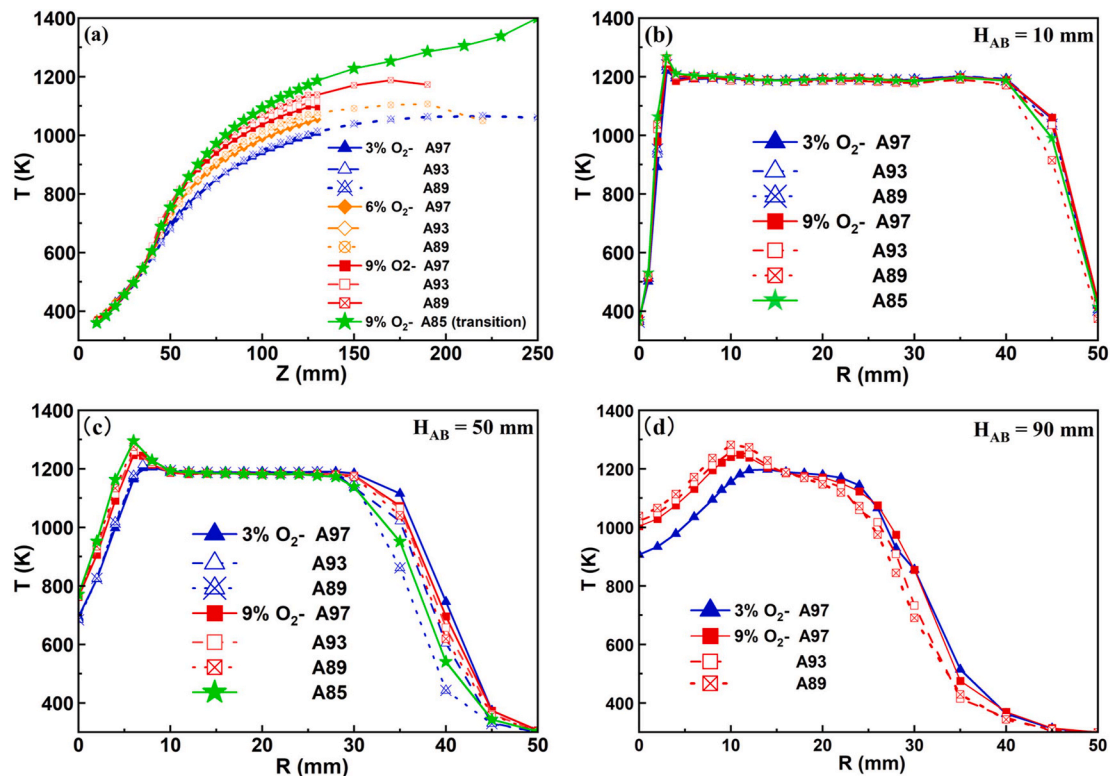


Fig. 5. Axial and radial temperature profiles of NH_3/H_2 flames with H_2 dilution greater than 1% (A97-A85), under co-flow oxygen concentrations of 3% , 6% and 9% . The uncertainty of the corrected thermocouple measurements is $\pm 2\%$.

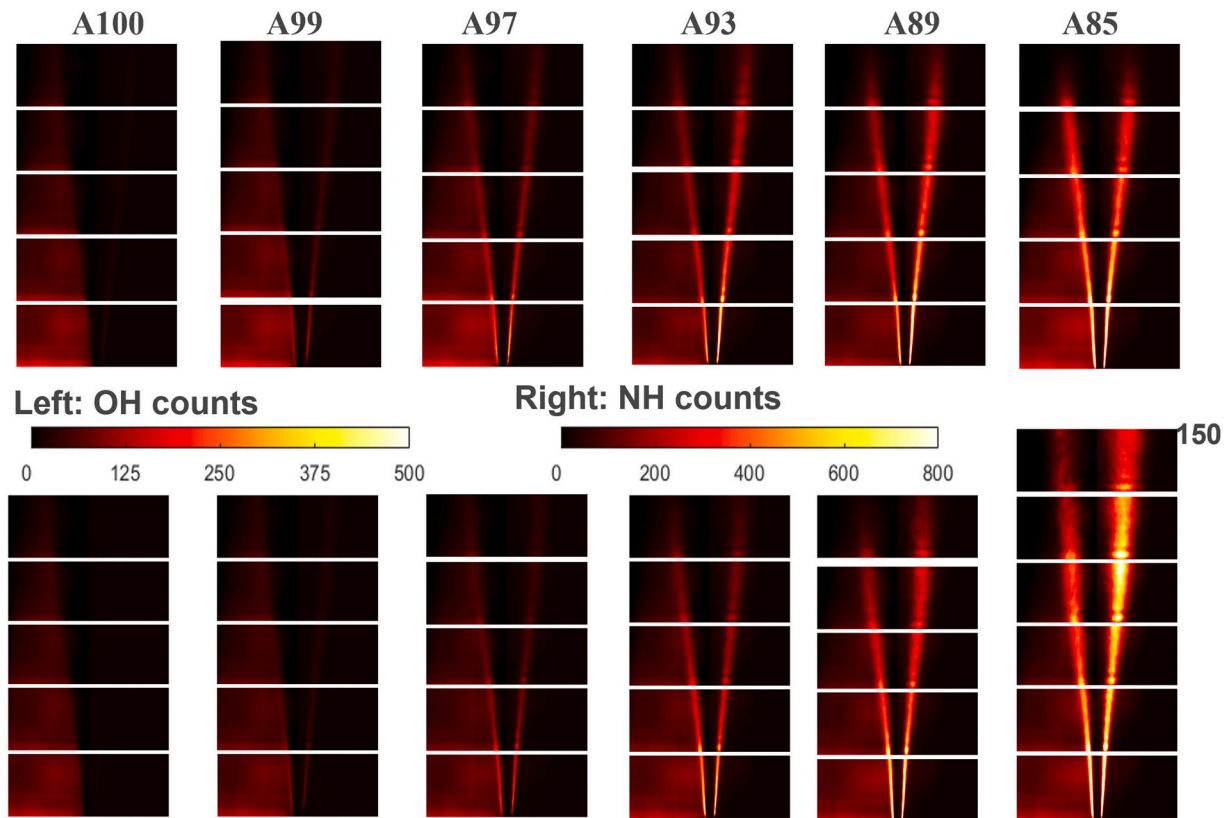


Fig. 6. Time-averaged NH-PLIF and OH-PLIF images of NH_3/H_2 flames at varying ammonia addition levels (A100-A85) under co-flow oxygen concentrations of 3 % (top) and 9 % (bottom). For each flame, the left half shows OH-PLIF, and the right half shows NH-PLIF.

condition, the flame exhibits characteristics of a standard turbulent flame.

Vertically, minor variations in flame structure are observed with different oxygen concentrations (9 % and 3 %), except in the A85O9 case, where the flame transits into a standard turbulent flame. While no such transition occurs at 3 % O_2 . This implies that lower oxygen concentrations necessitate higher hydrogen dilution in the jet to activate the reaction and trigger the transition. Additionally, the reaction zone broadens with decreasing NH_3/H_2 ratios, indicating enhanced combustion intensity.

To evaluate the spatial evolution of the reaction zone under varying hydrogen dilution levels at co-flow oxygen concentrations of 3 % and 9 %, the peak intensity normalized by the reaction zone area (Max/Area) was extracted from the time-averaged NH-PLIF measurements (as shown in Fig. 6) at three representative axial locations ($H_{AB} = 10, 50$, and 90 mm). At each axial plane, the radial intensity profile was obtained, and the local maximum was identified. The reaction zone was then defined by integrating the intensity between the two radial positions where the signal dropped to 5 % of the peak value of each side. This Max/Area metric serves as a measure of the spatial discreteness or concentration of the reaction zone.

As shown in Fig. 7, Max/Area exhibits a monotonic decrease with downstream distance across all conditions, reflecting the progressive broadening of the reaction zone in turbulent flames and the associated dispersion of scalar quantities. Taking the A100O9 case as an example, Max/Area decreases from approximately 0.80 at $H_{AB} = 10$ mm to 0.22 at $H_{AB} = 50$ mm and further to 0.10 at $H_{AB} = 90$ mm.

Near the flame root ($H_{AB} = 10$ mm), Max/Area increases from 0.68 to 0.80 as the co-flow oxygen concentration rises from 3 % to 9 % in the pure ammonia case, suggesting that low-oxygen environments favor the formation of a more distributed reaction zone. Additionally, increasing hydrogen dilution leads to a monotonic rise in Max/Area at this location

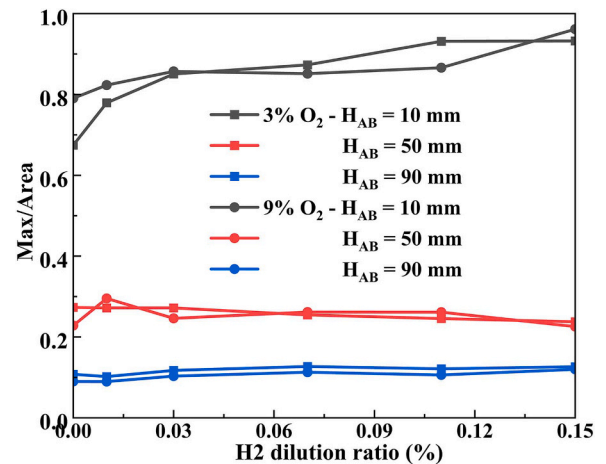


Fig. 7. Variation of normalized peak intensity (Max/Area) with different H_2 dilutions at $H_{AB} = 10, 50$ and 90 mm under co-flow oxygen concentrations of 3 % and 9 %.

for both oxygen levels, indicating the emergence of a more compact flame front, which is characteristic of non-premixed combustion.

At $H_{AB} = 50$ mm, Max/Area remains nearly constant (~ 0.25) across all test cases, showing minimal sensitivity to both hydrogen content and oxygen concentration. Further downstream at $H_{AB} = 90$ mm, Max/Area values converge toward ~ 0.10 , indicating a fully distributed reaction zone where the effects of local composition diminish.

Interestingly, despite the low Max/Area observed in the A85O9 case, the flame still exhibits features of standard turbulent combustion. This apparent inconsistency may arise from the interaction between the

flame and the relatively uniform co-flow thermal field at $H_{AB} = 50$ mm, which can suppress local gradients in temperature and radicals. As a result, the reaction zone is further broadened.

Presented in Fig. 8 are representative single-shot images of NH- and OH-PLIF for cases with hydrogen dilution ranging from 0 to 15 %, under a fixed co-flow oxygen concentration of 9 %. Numerical studies on ammonia flames reveal that the NH and OH radicals both serve as reliable indicators for characterizing the peak location and spatial distribution of the heat release rate (HRR) [40]. The simultaneous capture of these two radicals facilitates an in-depth characterization of the reaction zone structure in non-premixed combustion.

Both OH and NH signals reveal a vertically aligned reaction zone with low curvature and minimal structural variation across hydrogen dilution levels within the axial range of 0–50 mm. Although the reaction zone remains largely unchanged in this region, the NH-PLIF intensity increases markedly with hydrogen addition, indicating enhanced radical formation and reaction activity. This decoupling between flame structure and chemical reactivity suggests that hydrogen enrichment amplifies combustion intensity without altering the geometric characteristics of the reaction zone upstream. Further downstream ($H_{AB} > 100$ mm), the flame becomes progressively more wrinkled and

fragmented, particularly at higher hydrogen dilution levels (e.g., A85).

In the A97 case, the NH radical layer appears fragmented and discontinuous, with distinct axial and radial gaps. This indicates that under low hydrogen dilution, chain-branching reactions are significantly suppressed, and the reactive zone remains spatially confined. Local extinction events are a potential reason to explain why there exist discontinuous points. More hydrogen dilution significantly reduces the auto-ignition temperature and further increases the chemical reactivity of the NH_3/H_2 mixture. It is reasonable that the reaction zone with more hydrogen dilution raises the extinction rate, indicating improved flame stability due to more active chain-braining reactions

As the hydrogen dilution increases to A89, the reaction zone evolves into a more continuous, band-like structure. The NH layer becomes spatially coherent, with moderate wrinkling emerging along the reaction zone. Local bulges, fine-scale folds, and asymmetric flame edges become noticeable, reflecting enhanced radical production and improved flame sensitivity to turbulent eddies.

With further increases in hydrogen dilution to 15 %, the reaction zone becomes highly wrinkled and spatially extended, displaying pronounced multi-scale curvature and high-frequency surface perturbations. The reaction zone is no longer smooth but is dynamically

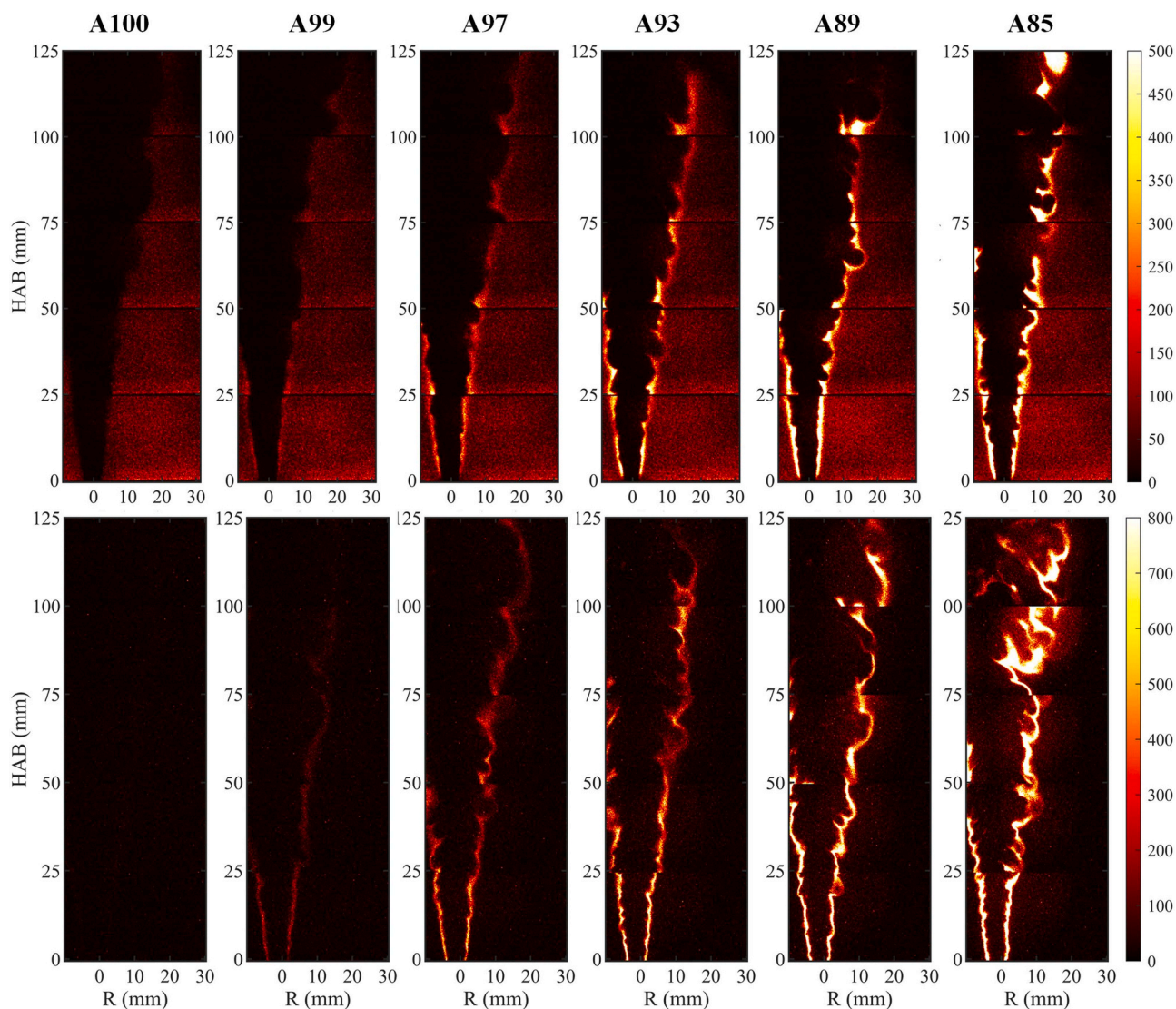


Fig. 8. Simultaneous single-shot images of OH-PLIF (top) and NH-PLIF (bottom) for various hydrogen dilution levels (A100, A99, A97, A93, A89, A85) at a fixed co-flow oxygen concentration of 9 %. Images of different heights in the same flame were taken at different instant of time. All the images share the same color bar range (0–1000).

stretched. These are characteristics of a wrinkled turbulent flame, where strong coupling between reaction zone strain and chemical kinetics occurs. The spatial extent of NH radicals significantly increases, indicating substantial expansion of the reactive reaction volume and intensified heat release.

Despite the non-premixed combustion mode, the reaction layer marked by NH remains very narrow, especially close to the nozzle, and the NH layer thickens slightly downstream due to the enhanced turbulence and higher temperatures. With the axial location greater than 100 mm, turbulence and entrainment of outer air cause the NH layer to wrinkle, fold, and widen strongly.

It is noteworthy that in the A85 case, significant flame intensification is observed in the axial range of 100 - 125 mm in NH-PLIF images. This enhancement is attributed to the enhanced oxygen availability induced by the entrainment of ambient air.

3.4. Combustion regime identification

Based on the auto-ignition temperature (T_{auto}) and the maximum temperature increase ($\Delta T_{\text{max}} = T_{\text{max}} - T_{\text{cof}}$), the hot co-flow non-premixed combustion modes in this study can be classified into two categories: unreacted combustion ($T_{\text{cof}}/T_{\text{auto}} < 1$ and $\Delta T_{\text{max}}/T_{\text{auto}} < 1$) and MILD combustion ($T_{\text{cof}}/T_{\text{auto}} > 1$ and $\Delta T_{\text{max}}/T_{\text{auto}} < 1$). T_{auto} and T_{max} were determined using a non-premixed opposed-flow flame in the ANSYS Chemkin-Pro 2023 [41]. The chemical kinetic mechanism of Zhang et al. [42] was employed for all the experimental conditions in Table 1.

Fig. 9 shows the combustion modes of NH_3/H_2 hot co-flow combustion with different H_2 dilution in the jet under all the investigated cases. The NH_3/H_2 flames mainly fall under the MILD combustion regime, characterized by excess oxygen concentration ranging from 3 % to 9 % (by mass), and hydrogen dilution changes from 1 % to 15 %. Pure NH_3 MILD combustion did not occur at any of the examined O_2 concentrations, underlining the critical role of T_{cof} in enabling pure ammonia combustion. However, it is observed that with 1 % H_2 dilution, the flame transitions into the MILD combustion regime regardless of the oxygen concentration. The auto-ignition temperature decreases significantly from approximately 1350 K for pure ammonia to ≈ 1150 K with 1 % hydrogen addition. This notable reduction highlights the promoting effect of hydrogen on the auto-ignition reactivity and facilitates the onset of MILD combustion conditions. The measured uncertainties were estimated to be ± 2 % for T_{cof} (1156–1204 K) and approximately ± 1 % for the H_2 mass flow. Propagation of these uncertainties into the 1D auto-ignition temperature calculations produced negligible variations,

confirming that the experimental co-flow temperature ($T_{\text{cof}} = 1180$ K) remains below the auto-ignition threshold (~ 1350 K at $\text{Y}_{\text{O}_2} = 9$ %).

Fig. 10 presents the changes in auto-ignition temperature and maximum temperature rise with hydrogen dilution in the jet. Both T_{auto} and ΔT_{max} vary monotonically with hydrogen dilution. T_{auto} exhibits a similar reduced trend, with the excess oxygen concentrations ranging from 3 % to 15 %. To be specific, T_{auto} drops from 1350 K to 1030 K with hydrogen dilution increases from 0 to 20 % (by vol.) under the co-flow temperature at ~ 1180 K. Notably, the rate of decrease becomes less significant at higher hydrogen dilution, indicating that even minor ammonia dissociation will help establish the MILD combustion at co-flow temperature as low as 1180 K. No peaks can be seen at pure ammonia condition since its autoignition temperature (~ 1350 K) exceeds the current co-flow temperature (~ 1180 K), which is consistent with the PLIF results in Section 3.3.

Importantly, a stable MILD flame is sustained with only 1 % hydrogen addition at a co-flow temperature of 1180 K, demonstrating that NH_3/H_2 mixtures readily develop MILD combustion when $T_{\text{in}} > T_{\text{aut},0}$, and that the criterion $\Delta T_{\text{max}} < T_{\text{aut},0}$ is more relaxed. Moreover, increasing the co-flow oxygen concentration to 15 % requires < 20 % hydrogen to satisfy $\Delta T_{\text{max}} < T_{\text{aut},0}$, indicating that the flame regime is governed by the interplay between fuel and oxidizer reactivity. Finally, it should be emphasized that temperature alone is insufficient to identify the MILD regime. Key indicators such as radical distribution, flame broadening, and NO suppression must also be considered to characterize MILD combustion accurately.

3.5. NO and N_2O emissions

One of the key advantages of MILD combustion is its potential to substantially reduce NO and N_2O emissions, owing to the lower peak temperature and spatially distributed reaction zone. In all experimental cases, the measured flame temperature remained below 1600 K. Combined with the reduced O_2 concentration in the co-flow, this effectively suppresses thermal-NOx pathway, leaving fuel-NOx as dominant formation mechanism. Fig. 11 presents the measured NO and N_2O emissions for the A100, A97, A97, A89 and A85 cases, all conducted with a fixed co-flow oxygen concentration of 9 % along the axial direction. As noted previously, the A10009 (pure NH_3) does not exhibit MILD-like characteristics. All flame cases exhibit a similar axial trend: NO formation is concentrated at relatively low temperatures, near the flame root, whereas N_2O is predominantly produced in the mid-to-downstream regions of the flame.

For NO emissions, the concentration increases along the axial direction, peaks, and then gradually decreases. The axial location of the NO peak correlates closely with the region of the maximum temperature gradient. At approximately 250 mm downstream, NO concentration in all four cases converges to similar levels, around 800 ppm. In contrast, N_2O , a potent greenhouse gas with a significantly higher global warming potential than CO_2 , exhibits a distinctly different behavior. Near the flame root, all cases show comparable N_2O concentrations (~ 50 ppm). However, as the hydrogen dilution level increases, the total N_2O emissions rise markedly. Specifically, the peak N_2O concentration increases from 90 ppm to 780 ppm as the hydrogen dilution ratio increases from 3 % to 15 %. The higher H_2 fraction promotes the production of H and OH radicals, which in turn enhances the $\text{NH}_3 \rightarrow \text{NH}_2 \rightarrow \text{NH}$ conversion pathway. This behavior is consistent with the intensified time-averaged NH-PLIF image presented in Section 3.3. The increased NH concentration in turn promotes N_2O formation through the $\text{NH} + \text{NO} \rightleftharpoons \text{N}_2\text{O} + \text{H}$ reaction [43], supported by other minor reaction pathways such as $\text{HNO} + \text{N} \rightleftharpoons \text{N}_2\text{O} + \text{H}$. The measured temperature range of 1200–1400 K corresponds to the conditions where N_2O formation is kinetically favored, while its thermal decomposition remains limited [42], leading to the observed accumulation of N_2O along the downstream axis with increasing hydrogen dilution. The close spatial correlation between the NH and OH radical distributions and the NO/ N_2O concentration profiles

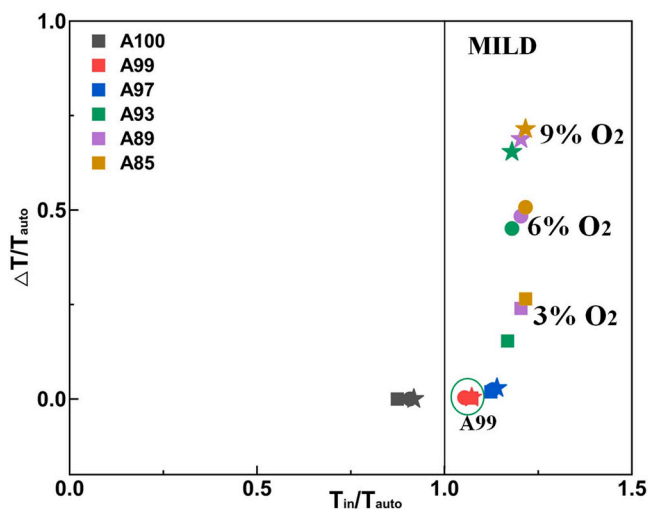


Fig. 9. Combustion modes of NH_3/H_2 jet flames with H_2 dilution ranging from 0 to 15 %, under the co-flow oxygen concentrations of 3 %, 6 %, and 9 %. Squares represent 3 % O_2 , circles represent 6 % O_2 , and stars represent 9 % O_2 .

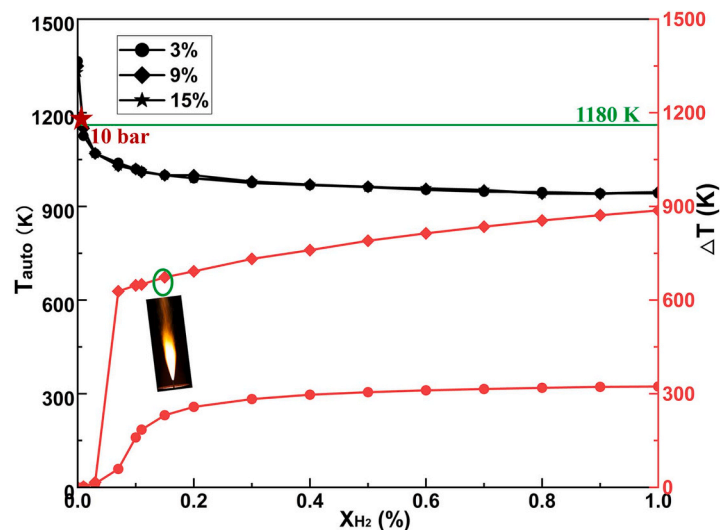


Fig. 10. Changes in auto-ignition temperature and maximum temperature rise of NH_3/H_2 flames with hydrogen dilution (0–100 % by vol.) in the jet, under co-flow oxygen concentrations of 3 %, 9 % and 15 %.

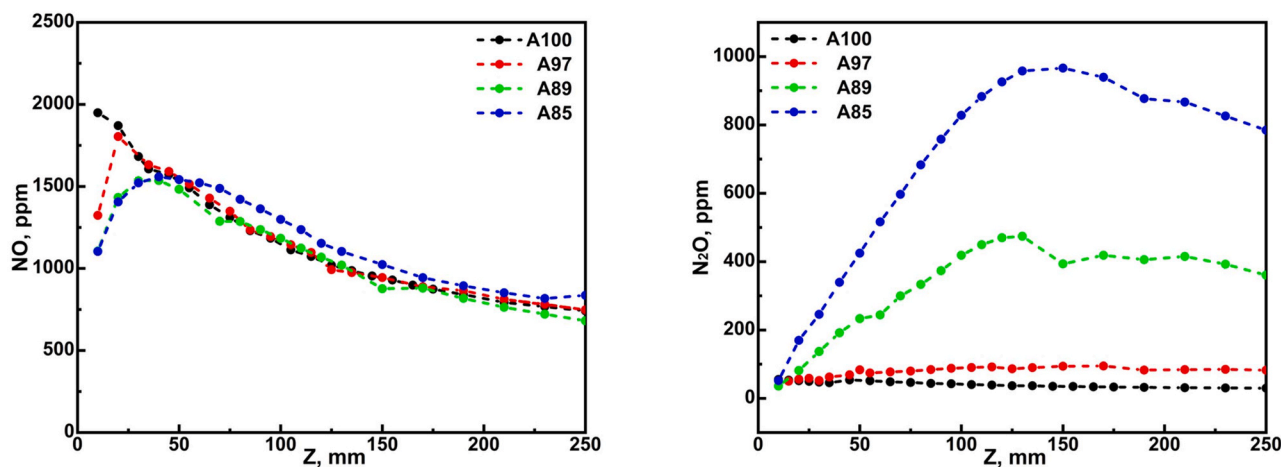


Fig. 11. NO and N_2O emissions along the axial direction for different hydrogen dilution levels (A100, A97, A89 and A85) in the jet, with the co-flow O_2 concentration at 9 %.

further confirms that NO formation proceeds primarily through fuel- NO_x chemistry ($\text{NH}_3 \rightarrow \text{NH}_2 \rightarrow \text{HNO} \rightarrow \text{NO}$) rather than thermal- NO_x . While hydrogen enrichment enhances flame stability and promotes distributed reaction behavior, it also introduces undesirable N_2O emissions. This trade-off highlights the delicate balance between improved reactivity and environmental performance.

Although cases A97, A89, and A85 satisfy conventional temperature-based criteria for MILD combustion, their NO and N_2O emissions remain relatively high. This observation aligns with the findings of Lewandowski et al. [33], who demonstrated that temperature-based MILD definitions alone are inadequate for ammonia-fueled systems due to complex fuel-nitrogen kinetics. They recommended additional dilution to suppress NO_x formation below 100 ppm. Similarly, Li et al. [44] showed that pollutant reduction in NH_3 MILD flames can be achieved under an equivalent ratio near unity, a co-flow temperature above 1500 K, and O_2 levels between 1–3 % which collectively suppress the HNO pathway and favor DeNO_x reactions that convert NO to N_2 . Other findings by Khamedov et al. [45] further revealed that the spatial distribution of reactive radicals such as NH_2 and O provides a more reliable indicator of distributed heat release than temperature alone. Building on these insights, a more robust classification of NH_3 -related MILD

combustion should incorporate additional criteria beyond temperature. These include radical-field distributions that identify spatially extended reaction zones, flame broadening indicative of volumetric reaction, and pollutant characteristics such as $\text{NO}_x/\text{N}_2\text{O}$ emissions.

4. Conclusions

This study experimentally investigated NH_3/H_2 non-premixed flames stabilized on a Jet-in-Hot-Co-flow (JHC) burner, with a particular focus on evaluating the feasibility and defining the boundary conditions for MILD combustion. Temperature profiles and PLIF measurements of NH and OH were acquired to characterize the onset and spatial evolution of the combustion process. Based on the temperature-based criteria, MILD combustion regimes were identified, and the associated N_2O and NO emissions were quantified.

All experiments were conducted over a range of co-flow oxygen concentrations (3 %–9 % by mass) and hydrogen volume fractions (0–15 %) with the fixed co-flow temperature at 1180 K.

The main conclusions are as follows:

- a) At 9 % oxygen concentration, 1 % H₂ dilution was sufficient to initiate a transition from non-burning to MILD combustion, whereas 15 % H₂ dilution led to the development of a standard turbulent flame. The latter transition was not observed at 3 % oxygen concentration, highlighting the critical influence of oxidizer on reactivity.
- b) 1 % H₂ dilution was sufficient to reduce the auto-ignition temperature of NH₃/H₂ below co-flow (1180 K), independent of the current oxygen concentration, indicating that minor NH₃ dissociation helps enable MILD combustion.
- c) NO emissions exhibited limited sensitivity to hydrogen dilution, with all the flames producing comparable values (~800 ppm) at 250 mm downstream of the jet. Irrespective of the initial condition. In contrast, N₂O levels were primarily formed in relatively higher-temperature regions and increased significantly with higher H₂ dilution, indicating optimize NH₃/H₂ blending for MILD combustion requires balancing enhanced reactivity against potential N₂O penalties.
- d) The present study was conducted at atmospheric pressure, which limits direct extrapolation to pressurized systems. Increasing the operating pressure is expected to lower the auto-ignition temperature of pure NH₃, thereby promoting earlier ignition and facilitating MILD combustion. Similarly, higher co-flow temperatures would help satisfy the ignition requirement for pure ammonia. Together, these effects would broaden the MILD operating window and enhance the applicability of the findings to practical burner conditions.

CRediT authorship contribution statement

Lele Ren: Writing – original draft, Methodology, Formal analysis, Data curation. **Adamu Alfazazi:** Writing – review & editing, Software, Methodology, Data curation. **Aurora Maffei:** Writing – review & editing, Data curation. **Sonu Kumar:** Writing – review & editing, Data curation. **Heinz Pitsch:** Writing – review & editing. **Bassam Dally:** Writing – review & editing, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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